

Resolving Incident-Induced Traffic Gridlock: Approximate Dynamic Programming vs. Large Language Models

Yishan Lu
ID:111062312
NTHU CS
sammy.yishan.lu@gmail.com

Borong Chen
ID:112062340
NTHU CS
yawaouter@gmail.com

Shaochieh Huang
ID:112062229
NTHU CS
zzzmax2004@gmail.com

Chiahsun Chang
ID:111062334
NTHU CS
a09001659560@gmail.com

Weifan Ching
ID:112062228
NTHU CS
venceching1015@gmail.com

Abstract—Resolving urban traffic gridlocks triggered by sudden incidents requires precise spatial reasoning and sequential decision-making. Traditional reactive methods, such as Max Pressure, often fail to prevent cascading congestion. Similarly, while formal planning-based approaches offer structured state-space models, they are primarily designed for normal traffic and struggle with extreme, highly coupled gridlock scenarios. Furthermore, Large Language Models (LLMs) may struggle to ensure physically feasible phase transitions and suffer from high latency in dynamic networks. In this work, we formulate traffic signal control under incident-induced congestion as a Decentralized Markov Decision Process (Dec-MDP) and simulate severe gridlock scenarios using the Simulation of Urban Mobility (SUMO). We propose a multi-agent Approximate Dynamic Programming (ADP) framework with linear value function approximation to derive effective control strategies. Each intersection agent is guided by a hybrid reward function that balances local queue conditions and global spatial-temporal incident features, enabling the learning of anticipatory policies. We expect the proposed approach to reduce Time-to-Recovery (TTR) while producing physically feasible control sequences, demonstrating the advantage of structured algorithmic methods over LLM-based strategies in constraint-driven environments.

Index Terms—Traffic gridlock, Approximate Dynamic Programming, Decentralized MDP, Multi-agent system, Traffic signal control.

I. INTRODUCTION

A. Motivation

In modern cities, traffic congestion has become a critical issue affecting both urban efficiency and quality of life. When a key road is suddenly blocked due to an incident, its capacity drops to zero, forcing vehicles to reroute or make U-turns, which disrupts traffic balance and propagates congestion across the network. This often leads to multiple intersections becoming mutually blocked, forming severe traffic gridlock that can paralyze the entire system. Therefore, rapidly restoring traffic flow after such incidents is a crucial challenge in Intelligent Transportation Systems.

B. Relevance

Resolving traffic gridlock is technically challenging, as it requires modeling dynamic traffic interactions while ensuring physically valid signal transitions. Existing methods have notable limitations. Max Pressure control, although effective in maximizing throughput, is inherently reactive and cannot prevent cascading congestion before it reaches a given intersection. Furthermore, while formal planning-based approaches (e.g., PDDL+) provide structured state-space search models, they are primarily designed for normal conditions and struggle to address extreme gridlock scenarios where intersections become highly coupled. Meanwhile, Large Language Models (LLMs), despite their recent success, may suffer from high latency and generate physically infeasible control actions in such structured environments.

C. Goal

To address these challenges, this study investigates how to coordinate traffic signals across multiple intersections through algorithmic design. We model each intersection as an independent agent within a Decentralized Markov Decision Process (Dec-MDP) and employ Approximate Dynamic Programming (ADP) guided by a hybrid reward function to learn a value function that captures both local queue conditions and global incident-related features. Using the Simulation of Urban Mobility (SUMO) environment, our goal is to restore all queue lengths to their pre-incident levels as quickly as possible.

D. Challenges

This problem presents several key challenges. The traffic system has a large state space, making exact solutions infeasible and requiring approximation methods, such as linear value function approximation, to serve as a practical and efficient baseline. In addition, decisions are highly interdependent across agents, creating a trade-off between local optimization and global stability. Finally, control decisions must be made

in real time under physical constraints, further increasing the complexity of the problem.

II. RELATED WORK

A. Max Pressure Control for Signalized Intersections

Max Pressure control, famously formalized by Varaiya, is a foundational algorithm designed to maximize network throughput and prevent congestion. Inspired by back-pressure routing in communication networks, the algorithm calculates the “pressure” of each signal phase. This pressure is defined as the difference between the number of vehicles waiting to enter the intersection (Q_{in}) and the queue length in the downstream receiving lanes (Q_{out}). At each decision interval, the agent automatically selects the signal phase that relieves the maximum pressure.

Limitation and Research Gap: While Max Pressure is mathematically proven to stabilize traffic networks without needing global Origin-Destination (OD) data, it is inherently a **strictly localized and reactive** strategy. In the event of a sudden accident, a Max Pressure agent will not alter its signal timing until the congestion (Q_{out}) physically backs up to its specific intersection. This delayed reaction allows vehicles to continuously pour into the bottleneck, inevitably leading to a cascading gridlock that paralyzes the grid.

Connection to Our Approach: Our proposed method retains Max Pressure’s foundational insight of utilizing real-time queue lengths for localized control, as reflected by the queue-based parameters ($\sum Q_d$ and $\max(Q_d)$) in our state representation. However, we overcome its strictly reactive limitation by formulating the problem as a Multi-Agent Markov Decision Process (MDP) solved via Approximate Dynamic Programming (ADP). Instead of greedily relieving immediate local pressure, our ADP model learns an anticipatory value function that integrates both local queue metrics and global spatial-temporal incident features (*dist* and *timeDiscrete*). This critical advancement enables our intersection agents to look beyond immediate downstream queues and proactively adjust signal phases, effectively preventing cascading gridlocks before the physical spillback reaches them.

B. Planning-based Traffic Signal Control

Beyond reactive control strategies such as Max Pressure, recent studies have explored planning-based approaches for traffic signal optimization. In particular, Percassi et al. formulate traffic signal control as a state-space search problem using PDDL+, where traffic dynamics and signal operations are explicitly modeled within a formal planning framework.

In this formulation, traffic conditions such as link occupancy and signal phases are represented as states, while signal timing adjustments are treated as actions. The objective is to find a sequence of actions that leads to desirable traffic flow conditions. To guide the search efficiently, domain-specific heuristics (e.g., hTSO) are introduced to estimate vehicle traversal times based on flow characteristics, enabling faster convergence to high-quality control strategies.

Limitation and Research Gap: Although planning-based approaches provide a structured way to model traffic dynamics and incorporate domain knowledge, they are primarily designed to optimize traffic flow efficiency under normal conditions. These methods often assume that congestion remains within manageable levels and do not explicitly address extreme scenarios such as gridlock, where queues exceed physical road capacity and interactions between intersections become highly coupled.

Connection to Our Approach: Inspired by planning-based formulations, our work also models traffic signal control as a state-space search problem. However, instead of focusing solely on throughput optimization, we explicitly target gridlock resolution under incident-induced congestion. Furthermore, rather than relying on predefined planning models such as PDDL+, we integrate simulation-driven state transitions from SUMO and design a heuristic tailored to estimating remaining road capacity. This allows our approach to better capture dynamic, large-scale traffic interactions and produce feasible control sequences in highly constrained environments.

III. PROBLEM FORMULATION

The traffic signal control problem under incident-induced congestion is modeled as a Decentralized Markov Decision Process (Dec-MDP). The road network is represented as a directed graph where each intersection $i \in \{1, 2, \dots, n\}$ is controlled by an independent learning agent.

A. System Model and Initial State

The traffic dynamics are simulated using SUMO. Prior to any disruption, the network operates under normal traffic demands, and the maximum acceptable queue length for each intersection i and direction d is recorded as the baseline threshold, denoted as $\bar{Q}_{i,d}$.

An episode is initiated at time step $t = 0$ (denoted by *timeDiscrete* = 0) when a traffic incident is triggered at a specific location, reducing the affected link’s capacity to zero and causing dynamic vehicle rerouting.

B. State Space

To support local value function approximation, the observation state S_i for agent i at decision step t is defined as a tuple comprising local traffic conditions and incident information:

$$S_i = (\{Q_{i,d}\}_{d \in \mathcal{D}}, D_i, dist_i, timeDiscrete) \quad (1)$$

where:

- $\mathcal{D} = \{N, S, E, W\}$ is the set of incoming directions.
- $Q_{i,d}$ is the real-time queue length at intersection i from direction d .
- $D_i \in \mathcal{D}$ represents the currently active green phase.
- $dist_i$ is the topological distance from intersection i to the incident location.
- *timeDiscrete* is a discretized scalar representing the elapsed time since the incident occurred.

The global state is the joint state $S = (S_1, \dots, S_n)$.

C. Action Space and Transition

Each agent i independently selects a discrete action $a_i \in \mathcal{D}$, indicating the direction to be granted the green light for the upcoming decision interval Δt (e.g., 10 seconds). When one direction receives the right-of-way, the other three are set to red.

The state transition $P(S'|S, A)$, where $A = (a_1, \dots, a_n)$ is the joint action, is determined by the SUMO simulator, which updates vehicle positions and queue lengths based on car-following and routing models.

D. Hybrid Reward Function

To balance local queue management with global network stability, we formulate a hybrid reward structure $R_i(S, a_i)$ for each agent i . It consists of a local queue penalty, a phase-switching penalty, and a global gridlock penalty:

$$R_i(S, a_i) = r_i^{local}(S_i) - \lambda \cdot \mathbb{I}(a_i \neq D_i) - P_{gridlock} \cdot \mathbb{I}_G(S) \quad (2)$$

where:

- $r_i^{local}(S_i) = -\sum_{d \in \mathcal{D}} \max(0, Q_{i,d} - \tilde{Q}_{i,d})$ penalizes the agent for local queues exceeding the baseline.
- λ is a constant penalty reflecting the capacity loss due to the clearance buffer (e.g., yellow/all-red time) when switching phases.
- $P_{gridlock}$ is a large constant penalty applied to all agents if the global gridlock condition indicator $\mathbb{I}_G(S)$ is triggered.

E. Objective and Episode Termination

The objective of each agent is to derive a policy that maximizes the expected cumulative hybrid reward, effectively minimizing the system's Time-to-Recovery (TTR) under the ongoing incident. Since the incident bottleneck is not physically restored during the simulation, an episode terminates under one of two conditions:

- 1) **Success (Stabilization):** The network successfully adapts to the incident and all local queues fall below a tolerable threshold ($\forall i, d : Q_{i,d} \leq \tau \cdot Q_{i,d}^{baseline}$), where $\tau \geq 1$ is a predefined tolerance factor (e.g., $\tau = 1.1$).
- 2) **Failure (Reset):** The global gridlock condition $\mathbb{I}_G(S) = 1$ is met, triggering an episode reset.

IV. PLANNED METHOD

A. Simulation Environment

In this project, we employ SUMO (Simulation of Urban Mobility) as the simulation environment. SUMO is an open-source traffic simulator capable of modeling vehicle flows, intersection queues and incident impacts. By leveraging SUMO, we can directly obtain state transitions and reward signals, which enables us to adopt Approximate Dynamic Programming (ADP).

B. MDP

Traffic signal control is a typical sequential decision-making problem: each intersection agent must continuously select signal phases under limited information, and these decisions affect future traffic states. This "state-action-transition-reward" structure naturally fits the Markov Decision Process (MDP) framework.

A more complete formulation has already been defined in the problem formulation section.

- **State:** Represents the real-time traffic condition of each intersection (queue length, incident location, incident duration)
- **Action:** Each intersection agent independently selects its signal phase.
- **Transition:** Provided by SUMO, reflecting vehicle flows and incident impacts.
- **Reward:** A hybrid design — each agent receives a local reward based on its own queue length, while a global penalty is applied if gridlock occurs.
- **Decision:** Requires an algorithm to derive the optimal policy that balances local responsiveness and global stability.

Thus, MDP provides a rigorous mathematical formulation, ensuring that strategies are theoretically grounded rather than relying solely on heuristic rules.

C. ADP

Since the traffic control state space is large, exact dynamic programming is infeasible. We adopt Approximate Dynamic Programming (ADP) with a linear value function approximation as a baseline, due to its simplicity and faster convergence. This allows us to quickly validate the framework and reward design under incident scenarios.

While linear approximation serves as an initial approach, we may try to further explore neural network function approximation. NN architectures allow non-linear interactions, ensuring that incident proximity and timing jointly influence decision-making. This extension would enable agents to better leverage incident-related features and improve generalization across diverse traffic scenarios. However, since training neural networks is computationally expensive and time-consuming, we prioritize linear approximation first to establish a practical and efficient baseline.

- **Approximate Value Function (Linear model):** Each agent i maintains its own local value function $V_i(S_i)$, approximated by a linear combination of input features such as queue length $Q_{d,i}$, incident distance $dist_i$, incident duration $timeDiscrete$, and other local traffic indicators. Although linear models are limited in capturing non-linear interactions, they provide a tractable starting point for training.
- **Update Method:** At each decision interval (e.g. every 10 seconds), each agent collects samples (S_i, A_i, R_i, S'_i) and updates its parameters using temporal-difference learning:

$$V_i(S_i) \leftarrow V_i(S_i) + \alpha(R_i + \gamma V_i(S'_i) - V_i(S_i))$$

where R_i is the hybrid reward (local queue penalty + global gridlock penalty).

- Policy Selection: For a given local state S_i , agent i computes the expected value of each possible action:

$$Q_i(S_i, A_i) = R_i(S_i, A_i) + \gamma V_i(S'_i)$$

Agent i then chooses the action that maximizes $Q_i(S_i, A_i)$.

- Gridlock Prevention Integration
 - Reward Penalty: Assign a large negative reward if global queue length exceeds a critical threshold.
 - Safety Policy: Enforce protective rules, e.g. prioritize directions with excessive queues.
 - Episode Reset: If gridlock still occurs, terminate and reset the episode

D. Execution Flow

The overall execution proceeds as follows:

- 1) Construct a stable road network in SUMO where vehicles can flow normally, and designate one road segment to be blocked in order to simulate an incident scenario.
- 2) Initialize agent parameters and weights for the linear value function $V_i(S_i)$.
- 3) At each decision interval, observe the current state S_i from SUMO (queue length, incident distance, incident duration, etc.).
- 4) For each possible action A_i , input (S_i, A_i) into the approximate value function to obtain the expected next-state value $V_i(S'_i)$, and then compute the expected action value:

$$Q_i(S_i, A_i) = R_i(S_i, A_i) + \gamma V_i(S'_i)$$

- 5) Select the action A_i^* that maximizes $Q_i(S_i, A_i)$.
- 6) Execute A_i^* in SUMO, obtain the next state S'_i and reward R_i .
- 7) Update the linear weights using temporal-difference learning:

$$V_i(S_i) \leftarrow V_i(S_i) + \alpha(R_i + \gamma V_i(S'_i) - V_i(S_i))$$

- 8) Repeat steps 3–7 until the episode ends; if gridlock occurs, apply penalties and reset the episode.
- 9) Over multiple episodes, the weights converge, yielding an improved traffic control policy that balances local responsiveness and global stability.

V. EXPECTED RESULTS

We aim to minimize the Time-to-Recovery (TTR) by maximizing cumulative rewards under the proposed framework. We expect our approach to produce efficient and feasible control strategies, demonstrating the advantage of structured algorithms over LLM-based methods in constraint-driven environments.